

WATER AUDIT

KATTAKKADA Legislative Assembly Constituency

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1. Background

'Vattatha Uravakkay Jalasamrudhi' is a project aimed at replenishing and rejuvenating water resources in Kattakada Legislative Assembly Constituency, in Thiruvananthapuram District. The project was launched on 22 March 2017, the World Water Day. As part of the project, detailed Management Action Plan was prepared with the help of various Government Departments and other agencies and many activities, trainings; workshops were carried out in the LAC. As a step forward, they wish to conduct water auditing of public buildings in the Kattakada constituency to convey the message to the public at large on water conservation and its judicious use. In addition, as a pilot project they wish to do it for eight public buildings in the LAC (Six Grama Panchayath Offices, One Block Panchayath Office and One Community Health Centre)

Adv. I B Sathish, MLA from Kattakada LAC sent a letter to CWRDM Executive Director to look into this matter. A proposal was prepared by CWRDM and was sent to Adv. I B Sathish, MLA, based on the above request and subsequent discussions the concerned Scientist had with various personnel involved in the project. Adv. I B Sathish, MLA sent the proposal to Kerala State Council for Science, Technology and Environment (KSCSTE) to take up the study. The Member Secretary, KSCSTE instructed the Executive Director, Centre for Water Resources Development and Management (KSCSTE-CWRDM) to conduct the study. As per directions from the Executive Director CWRDM, Scientist & officer in charge of Neyyattinkara Sub-centre of CWRDM visited these public buildings and conducted the study and this report was prepared.

2. Introduction

The development of water resources to man's benefit has been a fundamental factor in the evolution of civilizations throughout the history. Human populations continue to grow and shift as per the availability of quality water resources. It is essential that man utilize existing water re-

sources in the most careful, efficient manner. Water audits provide a rational, scientific framework that categorizes all water use in your system. It is a tool to overcome water shortage, leakage, wastage and losses.

With the help of water audit, we can identify and quantify the water uses, formulate steps, which can be taken to reduce water use and losses. Water audit and its analysis that can solve not only many water related problems but also saves precious resources and public money. Some water losses are unavoidable, and it is not worth the expense to try to eliminate every drop escaping your system. However, most of the loss that occurs in water systems can be managed by conducting a water audit. The results can help you prioritize steps to implement cost-effective water-saving measures.

Places that serve everyone in the community-parks, libraries, public buildings, markets, hospitals, plazas, playgrounds, sidewalks and other hang outs-are more important. Hospitals and office buildings require a large water volume for mechanical systems, once the systems have been determined, a water savings plan can be developed. In this context eight (8) Public buildings, six (6) Grama panchayath offices, one Block panchayath office and one Community

health center, located in Kattakada LAC were selected to conduct the water audit, which is presented in the following sections of this report.

3. Kattakada LAC

Kattakada is situated at the centre of four places: Trivandrum to the west, Neyyattinkara to the south, Neyyar Dam to the east and Nedumangad to the north. Trivandrum district has a climate that borders between tropical savanna climate and tropical monsoon climate. In a broad sense, it can be said that the district experience a tropical monsoon climate. The annual variation of mean air temperature at Trivandrum district is from 21o C to 34o C. The humidity is high and rises about 90% during the monsoon season. The average annual rainfall of the district is 2035mm. It is significant that Kattakkada area gets benefits of both monsoon - southwest monsoon and northeast monsoon.

Figure1: Kattakada LAC location map

As per Economic Review 2018, State Planning Board, Kerala, the net annual groundwater availability of the Trivandrum district is 296.24 MCM whereas the gross groundwater draft for all users of the district is 178.54 MCM, considering the domestic

requirement until 2025 the net ground water availability for future is 114.45 MCM. The stage of ground water development is 60.27%. As per Ground Water information booklet of Trivandrum district (Ministry of Wa-

ter Resources, Central Ground Water Board, Govt. of India) out of 12 blocks (prior to reorganisation of blocks), five are 'Semi critical' and nine are 'Safe'. Athiyannur, Nedumangad, Nemom, Parassala and Trivandrum blocks in Trivandrum district falls in semicritical category. Kattakkada LAC belongs to 'semi-critical category' as it is located in Nedumangad and Nemom blocks. The groundwater in Kattakkada area is generally potable except in few locations. The water is generally alkaline in nature and all the major constituents are within the permissible limit. The groundwater is suitable for domestic and drinking purposes.

4. Objectives

The major objectives of the study are:

- □ To Conduct water audit of 8 public buildings in Kattakkada LAC;
- □ To understand the water usage pattern and consumption levels at each offices;
- □ To reduce water wastage and its unnecessary use;
- □ To identify opportunities to make water use more efficient;
- □ To identify ways to conserve water; and
- □ To create awareness among users.

5. Public Buildings Selected for Study

Eight (8) public offices were selected to be included in this study. Out of these eight (8) offices, six (6) were Grama panchayath offices, of the remaining two (2) one was block panchayath office and the other one was a community health center. These buildings were spread across the LAC as shown in the Figure: 1 above. All these eight offices were visited to understand their water sources and its usage pattern. The basic details of these buildings collected during the visit are given below.

i. Pallichal Grama Panchayath Office

Location	Near Vedicachankoil, Pallichal, Kerala 08 0 26'00.9" N, 77 0 01'42.6" E
Water availability	Public water supply and well

ii. Vilavoorkal Grama Panchayath Office

Location	Near Pettayil, Kerala 695573 08 0 29'30.5" N, 77 0 01'10.9" E
Water availability	Public water supply

iii. Vilappil Grama Panchayath Office

Location	Aruvippuram, Near Peyad, Kerala 695573 08 0 30'51.2" N, 77 0 01'05.5" E
Water availability	Public water supply

iv. Malayinkeezh Grama Panchayath Office

Location	Papanamcode, Malayinkeezhu Rd, Malayinkeezhu, Kerala 695571 08 0 29'19.7" N, 77 0 02'15.1" E
Water availability	Public water supply

v. Maranalloor Grama Panchayath Office

Location	556A/X, Pongummoodu Punnavoor Rd, Maranalloor, Kerala 695512 08 0 28'09.4" N, 77 0 04'25.0" E
Water availability	Public water supply

vi. Kattakada Grama Panchayath Office

Location	College Rd, Kattakada, Thiruvananthapuram, Kerala 695572 08 0 30'27.4" N, 77 0 04'56.8" E
Water availability	Public water supply

vii. Nemom Block Panchayath Office

Location	Pongummoodu Rd, Malayinkeezhu, Kerala 695571 08 0 28'51.6" N, 77 0 02'46.6" E
Water availability	Well

viii. Malayinkeezh Community Health Centre

Location	Pongummoodu Rd, Malayinkeezhu, Kerala 695571 08 0 28'59.8" N, 77 0 02'41.7" E
Water availability	Public water supply, Well

6. Methodology

The methodology of water audit is the framework for categorizing and quantifying all water uses. American Water Works Association (AWWA) and International Water Association (IWA) developed a graphical way to think about the standard water balance in 2000, which is presented in Figure 2.

Figure 2: AWWA/IWA Water Balance

It is important to understand that the vertical height of each category represents a proportional amount of water. Thus, the height of the System Input category represents all water pumped by the system in a given time period. This amount of water can be broken down into two

additional categories, Authorized Use and Water Losses. This vertical height water measurement holds true across the entire standard water balance. Following identify important relationships just by glancing at the standard water balance.

- $\square$
- $\square$
- $\square$
- System Input = Authorized Use + Water Losses;
- Water Losses = Apparent Losses + Real Losses;
- Non-revenue Water
- $\square$ Apparent Losses
- = Water Losses + Unbilled Authorized Use.
- = Metering Inaccuracies + Unauthorized Use.

- $\square$ Real Losses = Leakage on Mains + Overflows on Storages + Leakage on Service Connections

To calculate the above-mentioned equations the below given five steps are to be followed:

- Source Evaluation.
- Calculation of Authorized Consumption.
- Evaluation of Apparent Losses.
- Evaluation of Real Losses.
- Performance Measurement.

The following steps were carried out to understand the water system of the selected eight public offices

- Preliminary site visit to the offices to understand the nature of water uses and the systems installed in the buildings
- Discussions with the administrative department on current situation and the past trends in water consumption, current sources, supply amount, source metering, distribution, storage, wastewater generation etc.
- Sample survey for finding quantity of water consumption.
- Records of water pumped to overhead tanks and water withdrawn from public water supply (recorded through meters) were collected.

The following issues were noted during the process:

- $\square$ Lack of proper inflow measurement system
- $\square$ Installed meters at the inlet points of public water supply are not operational
- $\square$ Categorywise/sectionwise metering systems are not in place
- $\square$ Non-availability of manpower to monitor daily water inflow/pumping
- $\square$ Pumping from wells even before the overhead tanks go empty
- $\square$ Combined use of Public water supply and water from well at the same time

Due to the above mentioned constraints it was decided to follow the first equation of water balance in this study, i.e., System Input = Authorized Use + Water Losses,

Therefore,

$$\text{Water Losses} = \text{System Input} - \text{Authorized Use}$$

7. Water Audit and Assessment

System Input & Authorized Use Assessment

The following activities were carried out to understand the system use and authorized use of the eight public buildings

- Visits to the buildings to understand the nature of water uses and the systems installed in the buildings;
- Discussions with the administrative department on current situation and the past trends in water consumption, current sources, supply amount, source metering, distribution, storage, wastewater generation etc.; and
- Sample survey for finding quantity of water consumption
- Records of water pumped to overhead tanks and water withdrawn from public water supply (recorded from water meter) were collected to estimate actual supply

As per the available details from each office, the system input and authorized use were calculated as given below for each office.

7.1 Pallichal Grama Panchayath Office

An Open-Well existing inside the office area is the only source of water for this threestorey building. The water from the well is pumped using a motor pump to an overhead storage tank having a capacity of 1000 Litres, placed on top of the third floor of the building.

System Input

Quantity of water pumped to overhead tank is obtained from survey conducted in the month of December 2018.

Average pumping time during the observed month was 20 Mins/day =
 20.5×40 System Input = 820 Litres/day

Authorized Use

In total, there are three toilets in the building with one each for all the floors. Four taps and two wash-basins were also identified to be used in the observed month. Ground floor has two water taps while first and second floor both provided with one water tap and one washbasin each. Water from the nearby well is pumped to an overhead storage tank of capacity 1000 Litres every morning except holidays.

This quantity helps to arrive at total consumption of water for each day. Number of plumbing points and average number of users (permanent

staffs and visitors) will give per capita per day water usage. This data is helpful for finding the water requirement and wastage of water. The practice of water usage among staffs will decide the extend of water wastage. The total staff strength at panchayat is 20. It is estimated that usually on a regular working day of the week around 35 people are usually present in the premises which includes staffs, peoples representatives, visitors, etc.

Average daily pumping time 20.4 minute

Daily drinking water consumption = $1.6 \times 35 = 56 \text{ L}$

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 6 \times 20 \times 90/100 = 270 \text{ L}$

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L

90 percent of staff present daily)

Flush water use in toilets by visitors = $0.5 \times 6 \times 15 = 45.0 \text{ L}$

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 315.0 L

Total time tap is opened on average day for face & hand washing by staff is $2 \times 1.0 \times 20 \times 90/100$

= 36 minutes

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes)(assuming 50 percent of staff using tap daily)

Water use in basins for face/hand washing by staff = $36 \times 4 = 144 \text{ L}$

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $15 \times 1 \times 20/60 = 5 \text{ minutes}$

)(assuming

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $5 \times 4 = 20 \text{ L}$

Total water use in basins for face/hand washing = 164 L

Total time tap is opened an average day for washing lunchbox/utensils by staff is $1 \times 2.0 \times 20 \times 90/100 \times 50/100 = 18 \text{ mins}$

= $18 \times 4 = 72 \text{ Litres}$

Total water use in floor mopping is approx 50 L/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 528.5 L per day.

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	56	8.5
2	Toilet flushing	315	47.9
3	Face/hand washing/lunch box washing	236	35.9
4	Mopping and cleaning	50	7.6
5	Total	657	100

Water Losses

Water Losses = System Input -Authorized Use

- = 820 -657
- = 163 Litres
- = 19.8 % of System Input
- = 24.8 % of Authorised Use

7.2 Vilavoorkal Grama Panchayath Office

They are using water from public water supply. As the water meter is concealed, water usage data is not available from the panchayat. In total there are 4 toilets in the building, which includes 2 in the Basement floor and one each for ground and first floors. Basement floor is provided with 4 water taps and 3 washbasins, Ground floor with one water tap and 3 washbasins and First floor has one water tap and one washbasin.

System Input

System input could not be calculated since the public water supply meter is concealed.

Authorized Use

The total staff strength at panchayat is 17 (15 permanent, 1 driver and 1 sweeper). It is estimated that usually on a regular working day of the

week around 47 people are usually present in the premises which includes staffs, peoples representatives, visitors, etc.

Daily drinking water consumption = $1.6 \times 47 = 75 \text{ L}$

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 5 \times 17 \times 90/100 = 191 \text{ L}$

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by visitors = $0.5 \times 5 \times 30 = 75 \text{ L}$

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 266 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 17 \times 90/100 \times 50/100 = 22.95 \text{ minutes}$

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 per-cent of staff using tap daily)

Water use in basins for face/hand washing by staff = $22.95 \times 5 = 115 \text{ L}$

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $15 \times 1 \times 30/60 = 7.5 \text{ minutes}$

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $5 \times 7.5 = 37.5 \text{ L}$

Total water use in basins for face/hand washing = 152.5 L

Total water use in floor mopping is approx 50 L/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 543.5 L per day.

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	75	13.8
2	Toilet flushing	266	48.94
3	Face/hand washing	152.5	28.06
4	Mopping and cleaning	50	9.2
5	Total	543.5	100

Water Losses

Water Losses = System Input - Authorized Use

Water losses could not be calculated since the system input was not estimated.

7.3 Vilappil Grama Panchayath Office

Newly constructed building with public water supply system. In total there are 10 toilets in the building, which includes 4 each for ground floor and first floor and 2 in the third floor. 10 water taps and 7 washbasins are also located in the building. The public water supply has no meter installed during the survey.

System Input

System input could not be calculated since the public water supply meter is not installed.

Authorized Use

The total staff strength at panchayat is 32. It is estimated that usually on a regular working day of the week around 72 people are usually present in the premises which includes staffs, peoples representatives, visitors, etc.

Daily drinking water consumption = $1.6 \times 72 = 115 \text{ L}$

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 5 \times 32 \times 90/100 = 360 \text{ L}$

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by visitors = $0.5 \times 5 \times 40 = 100 \text{ L}$

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 460 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 32 \times 90/100 \times 50/100 = 43.2 \text{ minutes}$

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 percent of staff using tap daily)

Water use in basins for face/hand washing by staff = $43.2 \times 5 = 216 \text{ L}$

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $15 \times 1 \times 40/60 = 10$ minutes

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $5 \times 10 = 50$ L

Total water use in basins for face/hand washing = 266 L

Total water use in floor mopping is approx 50 L/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 891 L per day.

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	115	12.91
2	Toilet flushing	460	51.63
3	Face/hand washing	266	29.85
4	Mopping and cleaning	50	5.61
5	Total	891	100

Water Losses

Water Losses = System Input -Authorized Use

Water losses could not be calculated since the system input was not estimated.

7.4 Malayinkeezhu Grama Panchayath Office

Primary source of water is public water supply. In total there are 4 toilets in the building, which includes both for men and women. Except for basement, all the other floors have two toilets each. In addition 4 taps and 8 washbasins are also in use. Ground floor and first floor each provided with 2 water taps each. Ground floor has 3 washbasins also for water usage. Basement floor is only provided with 5 washbasins. Water usage data is not available from panchayat.

System Input

Quantity of water pumped to overhead tank were obtained from survey conducted in the month of December 2018 and April 2019.

Average pumping time during the observed month was 10 Mins/day

= 10×40

= 400 Litres

Average use of water from public water supply is 1 KL/day

= 1000 Litres

Total system input

= 1400 Litres/day

Authorized Use

The total staff strength at panchayat is 40. It is estimated that usually on a regular working day of the week around 90 people are usually present in the premises which includes staffs, peoples representatives, visitors, etc.

Daily drinking water consumption = $1.6 \times 90 = 144$ L

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 5 \times 40 \times 90/100 = 450$ L

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by visitors = $0.5 \times 5 \times 50 = 125$ L

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 575 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 40 \times 90/100 \times 50/100 = 54$ minutes

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 percent of staff using tap daily)

Water use in basins for face/hand washing by staff = $54 \times 5 = 270$ L

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $15 \times 1 \times 50/60 = 12.5$ minutes

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $5 \times 12.5 = 62.5$ L

Total water use in basins for face/hand washing = 332.5 L

Total water use in floor mopping is approx 50 L/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 1101.5 L per day.

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	144	13.07
2	Toilet flushing	575	52.2
3	Face/hand washing	333	30.19
4	Mopping and cleaning	50	4.54
5	Total	1102	100

Water Losses

Water Losses = System Input -Authorized Use

- = 1400 -1102
- = 298 Litres
- = 21.2 % of System Input
- = 27.0 % of Authorised Use

7.5 Malayinkeezhu Community Health Centre

The office building receives majority of its water supply from a well. Since this is a Health centre essential buildings including Office section, OP building, Impatient block, Laboratory etc. are located in the compound. Average no of users for the month of December 2018 is 190 which includes permanent staffs, patients and visitors.

System Input

Quantity of water pumped to overhead tank were obtained from survey conducted in the month of December 2018 and April 2019.

Average pumping time during the observed month was 279 Mins/day

= 279 X 30

= 8370 Litres

Total system input

= 8370 Litres/day

Authorized Use

Walked through the entire hospital area to understand the nature of water uses and the systems installed in the buildings. In total there are 21 toilets in the building, which includes both for men and women. In which 12 toilets are located in Impatient block (3 in ground floor and 9 in first floor) and 6 toilets in new OP building. Office section has two toilets with washbasin for each. two water taps are also identified in the Office section. Remaining one toilet is placed outside the laboratory building. In total 47 water taps and 33 wash basins are also identified in the Health centre compound.

Water from the well is pumped directly into four overhead water tanks (each with capacity 1000) which are connected in series. average 4-5 hours pumping time is required to fill all the tanks.

The total staff strength at hospital is 76 (56 permanent and 20 provisional staffs). It is estimated that usually on a regular working day of the week around 190 people are usually present in the premises which includes staffs, patients, visitors, etc.

Average daily pumping time 279 minutes

Daily drinking water consumption = $1.6 \times 300 = 480$ L

(Per capita per day drinking water is 1.6 lpcd)

Flush water use by hospital staff = $2.5 \times 5 \times 76 \times 90/100 = 855$ L

(average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by patients & visitors = $2.5 \times 5 \times 114 = 1425$ L

(average daily flush in toilets by patient/visitor is 1.5)

Total flush water use in toilets = 1710 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 76 \times 90/100 \times 50/100 = 102.6$ minutes

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 percent of staff using tap daily)

Water use in basins for face/hand washing by hospital staff = $102.6 \times 5 = 513$ L

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by patients/visitors is $114 \times 2 \times 20/60$

= 76 minutes

(average daily frequency of hand washing per patient/visitor is 2 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $76 \times 5 = 380$ L

Total water use in basins for face/hand washing = 893 L

Total water use in Laboratory = 400 L

Total water use in washing utensil and Medical equipments = 500 L

Total water use in floor mopping is approx 500 L/day

Water use in Gardening is approx 1500 L/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 6553 L per day.

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	480	8.09
2	Toilet flushing	2280	45.51
3	Basin water use	893	23.77
4	Water use in Laboratory	400	5.32
5	Washing and Medical equipments	500	7.98
6	Mopping and cleaning	500	5.33
7	Gardening	1500	4
8	Total	6553	100

Water Losses

- Water Losses = System Input -Authorized Use
- = 8370-6553
- = 1817 Litres
- = 21.0 % of System Input
- = 28.0 % of Authorised Use

7.6 Nemom Block Panchayath Office Complex

Since this area is designated as an institutional area several other office areas including those of Block Panchayat building, Pattikajathi Vikasana office, Agriculture department. Ksheera Vikasana office, Finance department, Saksharatha Office, AEE office, Veterinary department, Canteen etc. are located in the compound.

System Input

Quantity of water pumped to overhead tank were obtained from survey conducted in the month of December 2018 and April 2019.

Average pumping time during the observed month was 112 Mins/day

= 112 X 35

= 3920 Litres

Total system input

= 3920 Litres/day

Authorized Use

In total there are 18 toilets in the building, which includes both for men and women. Except for Saksharatha department, Veterinary department, Rest room and Canteen. Rest room only has a water tap as a water connection.

Block panchayat office building has 5 toilets (2 inside and 3 outside) with 5 washbasins and 8 water taps. Pattikajathi vikasana office and Finance department have one toilet with a washbasin and a single water tap. 4 toilets, one washbasin and 4 water taps were installed for both Agriculture department and Ksheera vikasana department buildings. Two taps and two washbasins were in use in Canteen. 3 toilets are there in AEE office ground floor in addition to 4 water taps and one washbasin. Two washbasins and one water taps were located in AEE office first floor. Saksharatha department building has only one washbasin and water tap. Three washbasins and two water taps are located in Veterinary department building. Total number of water tap connections are 29 and that of washbasins are 18. Water pumped to a large storage tank constructed right side of the entrance gate daily except holidays. As per the available data from the Block panchayat it takes average 2 hours to fill the tank full.

The total staff strength at Block panchayat is 85 (31 in Panchayat building and 54 in other offices situated in the compound). It is estimated that usually on a regular working day of the week around 129 (85+44) people are

usually present in the premises which includes staffs, peoples representatives, visitors, etc. Average daily pumping time is 120 minutes. Water is pumped to an overhead tank of capacity 2000 L.

Daily drinking water consumption = $1.6 \times 129 = 206$ L

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 5 \times 85 \times 90/100 = 956$ L

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by visitors = $1.0 \times 5 \times 50 = 250$ L

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 1066 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 85 \times 90/100 \times 50/100 = 114.75$ minutes

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 percent of staff using tap daily)

Water use in basins for face/hand washing by staff = $114.75 \times 5 = 574$ L

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $44 \times 1 \times 20/60 = 14.67$ minutes

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $14.67 \times 5 = 73$ L

Total water use in basins for face/hand washing = 647 L

Total water use in canteen is approx 400 L/day

Total water use in floor mopping is approx 100 L/day

Water use in Gardening is approx 500 L/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 2919 L per day

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	206	9.39
2	Toilet flushing	1206	48.59
3	Hand washing	647	29.49

SL No	Activity	Water use in L/day	Percentage of total use
4	Mopping and cleaning	100	4.56
5	Canteen	400	6.83
6	Gardening	500	1.14
7	Total	3059	100

Water Losses

- Water Losses = System Input -Authorized Use
- = 3920-3059
- = 861 Litres
- = 22.0 % of System Input
- = 28.1 % of Authorised Use

7.7 Maranalloor Grama Panchayat Building

Krishibhavan building and Kalyanamandapam are neighboring Panchayat building in the same compound. Large quantity of water usage expected in kalyanamandapam during function days. 23 taps for washing and 5 toilets were installed. Wastage of water is also expected to be large. Krishibhavan block has two water taps and one washbasin. Panchayath building has 7 water taps (5 in ground floor and 2 in first floor) and a washbasin in the ground floor.

Primary source of water is public water supply. Panchayat and Krishibhavan buildings use water from public supply. Well located in the compound meets requirements for kalyanamandapam during function days.

System Input

Quantity of water used from public water supply were obtained from survey conducted in the month of December 2018 and April 2019.

Average use of water from public water supply is 9 KL/day

Total system input

Authorized Use

The total staff strength at panchayat and Krishibhavan is 38. It is estimated that usually on a regular working day of the week around 20 visi-

tors are present in the premises.

Daily drinking water consumption = $1.6 \times 58 = 93$ L

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 5 \times 38 \times 90/100 = 427.5$ L

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by visitors = $0.5 \times 5 \times 20 = 50$ L

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 477.5 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 38 \times 90/100 \times 50/100 = 51.3$ minutes

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 per-cent of staff using tap daily)

Water use in basins for face/hand washing by staff = $51.3 \times 5 = 256.5$ L

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $15 \times 1 \times 20/60 = 5$ minutes

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $5 \times 5 = 25$ L

Total water use in basins for face/hand washing = 281.5 L

Total water use in floor mopping is approx 100 L/day

= 9000 Litres

= 9000 Litres/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 952 L per day

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	93	9.77
2	Toilet flushing	477.5	50.16
3	Face/hand washing	281.5	29.57

SL No	Activity	Water use in L/day	Percentage of total use
4	Mopping and cleaning	100	10.5
5	Total	952	100

Authorised use for Grama Panchayath building alone is 952 Litres/day.

Water Losses

Water Losses = System Input -Authorized Use

Water losses could not be calculated since the water use of Kalayanaman-dapam could not be quantified and it varies as per people participation in each function.

7.8 Kattakada Grama Panchayath Building

Kattakada panchayat building located near Judicial First Class Magistrate Court and Panchayat building has AE office attached to it. In total there are 3 toilets which includes 2 in AE office. 4 washbasins and 2 water taps are also in use in AE office. Panchayat building ground floor has 2 water taps and one washbasin. First floor has one toilet, 3 washbasins and 2 water taps. The only source of water is Public water supply.

System Input

Quantity of water used from public water supply as per meter reading were obtained from survey conducted in the month of December 2018 and April 2019.

Average use of water from public water supply is 2.2 KL/day

Total system input

Authorized Use

The total staff strength at panchayat is 20. It is estimated that usually on a regular working day of the week around 30 visitors are present in the premises.

Daily drinking water consumption = $1.6 \times 50 = 80$ L

(Per capita per day drinking water is 1.6 lpcd)

Flush water use in toilets by staff = $2.5 \times 5 \times 20 \times 90/100 = 225$ L

(The average daily flush in toilets by staff is 2.5 & capacity of the urinal cistern is 5 L) (assuming 90 percent of staff present daily)

Flush water use in toilets by visitors = $0.5 \times 5 \times 30 = 75$ L

(The average daily flush in toilets by visitors is taken as 0.5)

Total flush water use in toilets = 300 L

Total time tap is opened on average day for hand washing by staff is $2 \times 1.5 \times 20 \times 90/100 \times 50/100 = 27$ minutes

(average daily frequency of hand washing per staff is 2 and average time the tap is opened while hand washing is 1.5 minutes) (assuming 50 percent of staff using tap daily)

Water use in basins for face/hand washing by staff = $27 \times 5 = 135$ L

(the average flow rate in the basins is around 5 litres per minute)

Total time tap is opened on average day for hand washing by visitors is $15 \times 1 \times 20/60 = 5$ minutes

(average daily frequency of hand washing per visitor is 1 and average time the tap is opened while hand washing is 20 seconds)

Water use in basins for face/hand washing by visitors = $5 \times 5 = 25$ L

Total water use in basins for face/hand washing = 160 L

Total water use in floor mopping is approx 50 L/day

Water usage for public Toilet complex

No of Taps = 06

No of Toilets = 06

= 2200 Litres

= 2200 Litres/day

Average no of Toilet uses per day

= 50

= 50×10

= 500 Litres/day

Average no of urinal uses per day

= 50

= 200×3

= 600 Litres/day

Total for Toilet complex

- = 1100 Litres/day

Therefore based on the above recordings, monitoring and calculation the total water consumption is 1690 L per day

SL No	Activity	Water use in L/day	Percentage of total use
1	Drinking	80	4.73
2	Toilet flushing	300	17.75
3	Face/hand washing	160	9.47
4	Mopping and cleaning	50	2.96
5	Toilet Complex	1100	65.09
6	Total	1690	100

Water Losses

- Water Losses = System Input -Authorized Use
- = 2200-1690
- = 510 Litres
- = 23.1 % of System Input
- = 30.2 % of Authorised Use

8. Conclusions and Recommendations

Conclusions

- None of the public buildings analyzed in the study have category wise water meters to understand category-wise usage. Water meters are available only for the main water supply line from KWA. Public buildings that have wells as main water source, water is pumped randomly as when required. For some buildings, the KWA water meter is not working/installed or is concealed.
- Water supply lines are very old for some of the buildings and line leakages would be present;
- The water leakages were not identified one by one and marked as part of the study;
- Since no category/section wise water meters are installed for these public buildings the authorised water usages were calculated based on the general criteria case to case and through surveys;
- The total system input water was calculated based on the surveys carried out during December 2018 and April 2019.

- Major component of water usage in these public buildings is for toilet/urinal flushing;
- On an average 20-30 % of the system input water is wasted by various means;

Recommendations

Strategies for reducing water consumption in general

- Install data-logging water meters.

Installing data-logging water meters on incoming water enables ongoing monitoring.

This helps to identify trends, usage patterns and leaks. The data loggers are usually sourced from your water retailer.

- Check for leaks

A significant amount of infrastructure is ageing and prone to leaks. As much of it is underground, leaks are not always easily identified. Data-logging meters at the boundary are useful because out-of-hours data (when water use is expected to be very low) becomes readily available.

- Install sub-metering

Metering of separate buildings and key processes enables usage and trends to be monitored throughout the site. The use of meters that are connected to a building management system (BMS) enables easy data collection.

- Install flow restrictors

A number of fixtures (for example, hand basins, sinks and wash troughs) have unregulated water flows that may lead to higher use. Installing flow restrictors with

balanced pressure is generally a cost-effective way to reduce water use without affecting functional requirements.

• Install dual-flush toilets

Replace old toilet with a low flush toilet (6 litres per flush), a high efficiency model (4.8 litres per flush), or a dual model (3 or 6 litres per flush). Older toilet models can use 13 to 26 litres of water per flush.

Replacing an old toilet with an ultra-low volume (ULV) 1.6 gallon flush model represents a 70% savings in water and will cut indoor water use by about 30%. Alternatively, consider purchasing a dual flush toilet or installing a dual flush converter that turns a standard toilet into a dual flush

toilet, saving an average family 15,000 gallons of water each year. More water can be used when it's needed.

- Reclaim or recycle water
- If possible replace old toilet with a dry toilet, which does not use any water for flushing
- Replumb old toilet to use rainwater for flushing
- Collecting rainwater will meet the demand for gardening purpose for times of low rainfall
- Install water tap aerators or more efficient fixtures. Aerators for hand/utensils washing can reduce the water consumption by 80-90% as per manufacturers